

Quiz 1

1. What does axis=1 mean in NumPy operations like np.max(x, axis=1)?

- (a) Find maximum column-wise
- (b) **Find maximum row-wise**
- (c) Flatten the array first
- (d) Find overall maximum value

2. What is the main difference between a Python list and a tuple?

- (a) Lists are faster than tuples
- (b) Tuples can store only numbers
- (c) **Lists are mutable, tuples are immutable**
- (d) Tuples use square brackets

3. What does the following Pandas code do?

```
df[(df["Category"] == "Electronics") &  
   (df["Country"] == "USA")]
```

- (a) Selects rows where either condition is true
- (b) Deletes rows from the DataFrame
- (c) **Selects rows where both conditions are true**
- (d) Sorts the DataFrame

4. Which Matplotlib function is used to display a graph on the screen?

- (a) plt.save()
- (b) plt.figure()
- (c) **plt.show()**
- (d) plt.display()

5. What is broadcasting in NumPy?

- (a) Sending arrays over a network
- (b) **Automatically expanding arrays of compatible shapes during operations**
- (c) Copying arrays into lists
- (d) Sorting multidimensional arrays

